

# Literature Status: Real Two-Dimensional $\ell_p$ Numerical Index

FA/Banach run `fa_banach_001`

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## Status

This is a literature-already-answered packet, not an original proof from the run. Problem 2 in Kadets–Martin–Paya, arXiv:math/0605781, asks whether the numerical index of the real two-dimensional space  $\ell_p^2$  is

$$\sup_{t \in [0,1]} \frac{|t^{p-1} - t|}{1 + t^p} \quad (1 < p < \infty).$$

Chiclana, arXiv:2606.01675, answers this affirmatively by proving that  $n(\ell_p^2) = v(J)$  for every  $p \geq 1$ , where

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

## Original Problem

The original source is:

V. Kadets, M. Martin, R. Paya, *Recent progress and open questions on the numerical index of Banach spaces*, arXiv:math/0605781.

The paper states that, in the real case, the displayed supremum above is the numerical radius of the norm-one operator  $J$  on  $\ell_p^2$ , and asks whether this upper bound is always the numerical index.

the closed convex hull of a set  $B \subseteq X$  and we denote by  $\text{ext}(C)$  the set of extreme points of a convex set  $C \subseteq X$ . A subset  $A$  of  $B_{X^*}$  is said to be *norming* (for  $X$ ) if

$$\|x\| = \sup\{|x^*(x)| : x^* \in A\} \quad (x \in X)$$

or, equivalently, if  $B_{X^*} = \overline{\text{co}}^{w^*}(\mathbb{T}A)$  (Hahn-Banach Theorem). Finally, for  $1 \leq p \leq \infty$ , we write  $\ell_p^m$  to denote the normed space  $\mathbb{K}^m$  endowed with the usual  $p$ -norm, and we write  $X \oplus_p Y$  to denote the  $\ell_p$ -direct sum of the spaces  $X$  and  $Y$ .

## 1. COMPUTING THE NUMERICAL INDEX

In view of the examples given in the introduction, the most important family of classical Banach spaces (in the sense of H. Lacey [47]) whose numerical indices remain unknown is the family of  $L_p$  spaces when  $p \neq 1, 2, \infty$ . This is actually one of the most intriguing open problems in the field but, very recently, E. Ed-Dari and M. Khamsi [21, 22] have made some progress. We summarize their results in the following statement and use it to motivate some conjectures.

**Theorem 1** ([21, 22]). *Let  $1 \leq p \leq \infty$  be fixed. Then,*

- (a)  $n(L_p[0, 1]) = n(\ell_p) = \inf\{n(\ell_p^m) : m \in \mathbb{N}\}$ , and the sequence  $\{n(\ell_p^m)\}_{m \in \mathbb{N}}$  is decreasing.
- (b)  $n(L_p(\mu)) \geq n(\ell_p)$  for every positive measure  $\mu$ .
- (c) In the real case,

$$\frac{1}{2}M_p \leq n(\ell_p^2) \leq M_p, \quad \text{where } M_p = \sup_{t \in [0, 1]} \frac{|t^{p-1} - t|}{1 + t^p}.$$

When  $p \neq 2$ , it is known that  $n(\ell_p^m) > 0$  for every  $m \geq 2$ , and also that  $v(T) > 0$  for every non-null  $T \in L(\ell_p)$  (see [22] Theorem 2.3 and subsequent remark), but we do not know if  $n(\ell_p) > 0$ . Observe that a positive answer to this question implies, thanks to (b) above, that  $n(L_p(\mu)) > 0$  for every positive measure  $\mu$ .

**Problem 1.** *Is  $n(\ell_p)$  positive for every  $p \neq 2$ ?*

With respect to item (c) in the above theorem, let us explain the meaning of the number  $M_p$ . It can be deduced from [20] §3 that, given an operator  $T \in L(\ell_p^2)$  represented by the matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , one has

$$(2) \quad v(T) = \max \left\{ \max_{\substack{t \in [0, 1] \\ z \in \mathbb{T}}} \frac{|a + d t^p + z b t + \bar{z} c t^{p-1}|}{1 + t^p}, \max_{\substack{t \in [0, 1] \\ z \in \mathbb{T}}} \frac{|d + a t^p + \bar{z} c t + z b t^{p-1}|}{1 + t^p} \right\}.$$

It follows that  $M_p$  is equal to the numerical radius of the norm-one operator  $U \equiv \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  in  $L(\ell_p^2)$  (real case), so  $n(\ell_p^2) \leq M_p$ . For  $p = 2$ , the operator  $U$  has minimum numerical radius, namely 0. We may ask if  $U$  is also the norm-one operator with minimum numerical radius for all the real spaces  $\ell_p^2$ .

**Problem 2.** *Is it true that, in the real case,  $n(\ell_p^2) = \sup_{t \in [0, 1]} \frac{|t^{p-1} - t|}{1 + t^p}$  for every  $1 < p < \infty$ ?*

In the complex case, the operator  $U$  acting on  $\ell_2^2$  satisfies  $v(U) = \|U\|$  (take  $z = i$  and  $t = 1$  in [2]) and, therefore, its numerical radius is not the minimum. Actually, one has

$$n(\ell_2^2) = \frac{1}{2} = v(S),$$

## Later Answer

The supporting source is:

R. Chiclana, *The Numerical Index of Two-Dimensional Real  $\ell_p$  Spaces*, arXiv:2606.01675.

The abstract states that the paper computes the numerical index of two-dimensional real  $\ell_p$ -spaces for all  $p \geq 1$  and confirms the conjectured formula in the two-dimensional real case. Theorem 1.1 states:

$$n(\ell_p^2) = v(J) \quad (p \geq 1).$$

# The Numerical Index of Two-Dimensional Real $\ell_p$ Spaces

Rafael Chiclana\*

## Abstract

The computation of the numerical index of classical Banach spaces is one of the original problems in the theory. In this paper, we compute the numerical index of two-dimensional real  $\ell_p$ -spaces for all  $p \geq 1$ . More precisely, we prove that

$$n(\ell_p^2) = v(J), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

confirming the conjectured formula in the two-dimensional real case.

## 1 Introduction

The numerical index is a classical isometric invariant of Banach spaces, introduced to quantify the relation between the norm and the numerical radius of bounded linear operators. Let  $X$  be a Banach space over the real field, let  $X^*$  denote its dual space, and let  $\mathcal{L}(X)$  denote the space of all bounded linear operators from  $X$  into itself. For  $T \in \mathcal{L}(X)$ , the numerical radius of  $T$  is defined by

$$v(T) := \sup\{|x^*(Tx)| : x \in X, x^* \in X^*, \|x\| = \|x^*\| = x^*(x) = 1\}.$$

The numerical radius is a seminorm on  $\mathcal{L}(X)$  and satisfies  $v(T) \leq \|T\|$  for every  $T \in \mathcal{L}(X)$ . The numerical index of  $X$  is the number

$$n(X) := \inf\{v(T) : T \in \mathcal{L}(X), \|T\| = 1\}.$$

The concept was introduced in the seminal work of Duncan, McGregor, Pryce and White [3], after a question of Lumer, and was developed systematically in the monographs of Bonsall and Duncan [1, 2]. One of the original problems in the theory was to compute the numerical index of classical  $L_p$ -spaces. In the real case, this problem remains open for  $1 < p < \infty$ ,  $p \neq 2$ . In the two-dimensional case, it was conjectured that the numerical index of  $\ell_p^2$  is attained by the rotation by  $90^\circ$ , which corresponds to the operator

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

In this paper, we prove the conjecture.

**Theorem 1.1.** *For every  $p \geq 1$ ,*

$$n(\ell_p^2) = v(J).$$

The cases  $p = 1$  and  $p = 2$  are classical. Moreover, as recalled in [6, Section 1], the range  $1 < p < 2$  reduces by duality to the range  $p > 2$ . Thus, in the proof, we always work under the assumption  $p > 2$ .

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## Verification Notes

The supporting paper is a separate later paper. It identifies the same two-dimensional real  $\ell_p$  conjecture by formula and proves it. In the text inspected here, the supporting paper does not appear to cite arXiv:math/0605781 directly; it cites later partial-progress papers and describes the same conjecture.

The packet does not address the neighboring 2006 questions about positivity of  $n(\ell_p)$ , equality  $n(\ell_p) = n(\ell_p^2)$ , or the complex two-dimensional formula.

## Search Bounds

The registry and solution index were checked for arXiv:math/0605781, arXiv:2606.01675, Chiclana, and the real two-dimensional  $\ell_p$  formula. The original and supporting arXiv PDFs were downloaded and inspected. No broader web, MathSciNet, or Zentralblatt search was performed.

## Review Recommendation

Verify the two displayed links in the transfer: the original source's identity between the supremum and  $v(J)$ , and the supporting theorem  $n(\ell_p^2) = v(J)$ . No other mathematical input is used.